

Temporally Constrained Segmentation Enables Robust Quantification of Scratch-Wound Healing Dynamics

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Abstract

Objective: To develop and validate a fully automated, training-free computational pipeline for scratch-wound quantification that enforces a per-column monotonic closure constraint, eliminating the non-biological area fluctuations produced by frame-independent segmentation.

Materials and Methods: A training-free pipeline segments phase-contrast time-lapse images using local variance-based texture analysis with RANSAC outlier rejection and Savitzky–Golay boundary smoothing, operating in $O(HW)$ per frame. A per-column Kalman filter with a hard monotonic safety net enforces strictly non-increasing boundary evolution. The pipeline was evaluated on 134 phase-contrast images from two independent experiments and two cell lines (iMC ISOR544C and iMC MUTR544C) across three culture conditions (ALK5 inhibitor at 0.1 mM, DMSO vehicle, and media control), used as a test bed to assess whether the pipeline resolves condition-dependent closure dynamics.

Results: Expert review confirmed 98.4% detection accuracy across 124 wound-bearing image. The automated quality-control module achieved a sensitivity of 1.00 for wound detection (NPV = 1.00; Cohen’s κ = 0.90 against expert verdicts). The temporal constraint yielded strictly non-increasing area trajectories by construction, with a monotonic-constraint across all approved trajectories. A linear mixed-effects model fit to the resulting edge-displacement trajectories detected a significant closure-over-time effect (β = 0.88 px/frame, $p < 10^{-12}$), demonstrating that the pipeline is able to capture differences between the experiments.

Conclusion: The proposed pipeline offers a training-free of supervised deep-learning segmenter or manual parameter tuning; the temporal constraint removes the non-biological area fluctuations that accumulate when frames are processed independently. The resulting trajectories are physically

consistent and structured for post hoc statistical analysis. The implementation is released in GitHub [1].

Keywords: scratch-wound assay, automated segmentation, monotonic closure constraint, local-variance texture analysis, Kalman filter, temporal constraint, training-free image analysis

Introduction

The scratch-wound assay is a widely used *in vitro* method for studying collective cell migration. By creating a cell-free gap within a confluent cell monolayer, time-lapse microscopy captures the progressive closure of the wound as cells at the wound margin proliferate and migrate inward [2–8]. Cell migration governs tissue repair, morphogenesis, and tumor invasion, making reliable quantification of wound closure dynamics a direct readout of migration capacity across pharmacological screening, developmental biology, and oncology.

Despite the assay’s conceptual simplicity, quantification remains a significant bottleneck. Manual measurements are labor-intensive, introduce inter-observer variability, and do not scale efficiently to large experiments. To address these limitations, several computational tools have been developed, including TScratch [9], ImageJ plugins [10], the Robust Quantitative Scratch Assay [11], AIM [12], CellProfiler pipelines [13], and more recently level-set methods with spatial sector analysis [14].

Scratch-wound experiments are inherently temporal, generating image sequences that capture the continuous progression of wound closure. While existing approaches have substantially improved throughput and reproducibility, segmentation is often performed on individual frames, with temporal information incorporated primarily during downstream analysis. Integrating temporal information directly into the segmentation stage enforces physical constraints on boundary evolution, eliminating the non-biological area fluctuations that accumulate when each frame is analyzed in isolation. Temporal constraints have previously been explored in related domains such as cell tracking [15–17] and video segmentation [18].

In this study, we propose an automated end-to-end pipeline for scratch-wound analysis that combines texture-based wound detection with temporal integration across image sequences. The pipeline uses local variance information to distinguish the wound channel from the surrounding cellular monolayer in phase-contrast images and incorporates temporal continuity to generate

consistent wound-closure trajectories [19, 20]. The pipeline additionally provides automated quality control, parallel processing of image sequences, and extraction of quantitative wound-closure metrics including wound area, closure speed, edge displacement, and closure fraction. The pipeline is fully automated and requires neither training data nor user-defined threshold tuning. To our knowledge, this is the first application of a hard monotonic closure constraint to scratch-wound segmentation.

We evaluated the pipeline using a standard scratch-wound migration assay in immortalized skeletal muscle cells [21, 22] imaged under three culture conditions: an ALK5i, a DMSO vehicle control, and a media-only control. These conditions provide a controlled test bed for assessing performance across distinct wound-closure trajectories.

Materials and Methods

Cell Culture and Experimental Design

The evaluation dataset comprises two cell lines (iMC ISOR544C and iMC MUTR544C) across three culture conditions, acquired in two independent experiments serving as biological replicates (EXP1, $n = 62$ images; EXP2, $n = 72$ images), for a total of 134 baseline ($t = 0$) wound regions. The three conditions are an ALK5 inhibitor at 0.1 mM (a selective TGF- β receptor type I antagonist), a DMSO vehicle control, and an untreated media reference used solely as a test bed to probe whether the pipeline resolves condition-dependent closure dynamics.

Confluent monolayers were wounded using a standardized mechanical scratch assay. Treatment media were applied immediately after wounding. Time-lapse imaging began at $t = 0$ (baseline, fully open wound) and continued for 25 hours, with images acquired at 1-hour intervals, yielding 25 timepoints per trajectory. Images were captured at a resolution of approximately 900×735 pixels per field of view.

iMCs were seeded into 96-well plates at a density of 5×10^4 cells per well in media containing either DMSO or the indicated drug conditions. After overnight incubation, wounds were created in the cell monolayers using the Incucyte[®] 96-Well WoundMaker tool. Cells were washed twice with PBS, after which conditioned media were reapplied. Images were acquired immediately after wounding and every hour for up to 24 hours using the Incucyte[®] Live-Cell Analysis System with the whole-well (4 $\times$) setting.

Automated Wound Segmentation

The quantification pipeline operates in two stages: a single-frame detector that estimates wound boundaries independently for each image, followed by a temporal integration stage that enforces a monotonic closure constraint across the time series. The resulting boundary estimates are converted into wound masks and quantitative measurements.

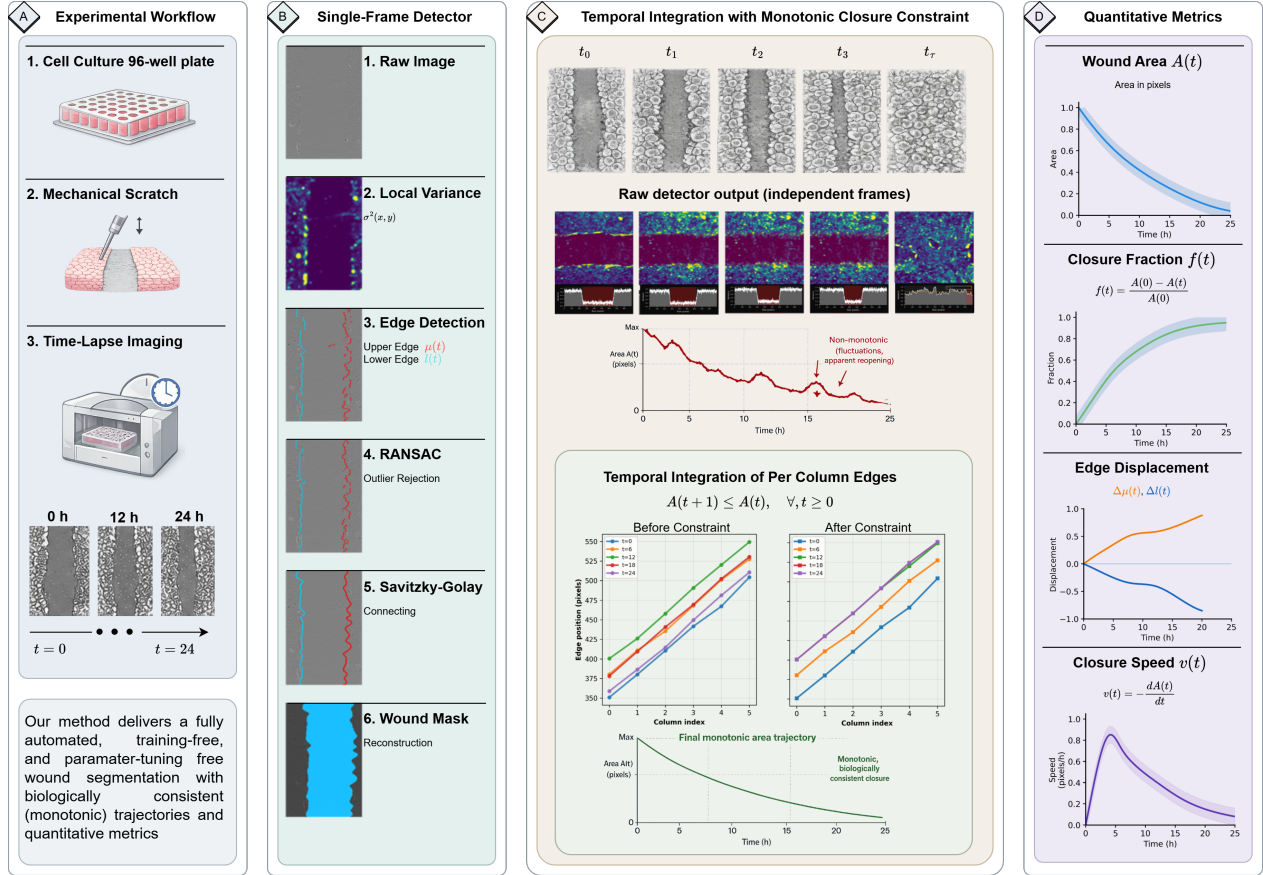


Figure 1: Automated scratch-wound quantification pipeline with monotonic temporal integration. (A) Experimental workflow. Confluent cell monolayers cultured in 96-well plates are wounded using a standardized mechanical scratch assay and imaged by phase-contrast time-lapse microscopy. Sequential images capture progressive wound closure from baseline to the final timepoint. (B) Single-frame wound detection. Each frame is independently segmented using local variance-based texture analysis to distinguish the wound channel from the surrounding cellular monolayer. Candidate wound boundaries are refined through edge detection, RANSAC-based outlier rejection, and Savitzky–Golay smoothing before reconstruction of the wound mask. (C) Temporal integration with monotonic closure constraint. Independent frame-wise segmentations produce wound-area trajectories, wound boundaries are modeled independently for each image column using a Kalman-filter framework with a hard monotonic constraint, ensuring that wound area cannot increase over time ($A(t+1) \leq A(t)$). This generates temporally consistent wound trajectories and strictly non-increasing area profiles. (D) Quantitative metrics derived from the final segmentation masks, including wound area ($A(t)$), closure fraction ($f(t)$), edge displacement ($\Delta u(t)$ and $\Delta \ell(t)$), and closure speed ($v(t) = -dA(t)/dt$). The fully automated, training-free pipeline enables robust extraction of wound-healing dynamics without user-defined threshold tuning while preserving biologically consistent temporal evolution.

Problem Formulation

Let $I(x, y, t) \in [0, 1]$ denote the intensity-normalized grayscale image at spatial position (x, y) and discrete timepoint $t \in \{0, 1, \dots, T\}$. The image has dimensions $H \times W$ (rows $\times$ columns). We seek a binary wound mask:

$$M(x, y, t) = \begin{cases} 1 & \text{if pixel } (x, y) \text{ is wound at time } t \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

and per-column edge functions:

$$u(x, t) = \min\{y : M(x, y, t) = 1\} \quad (\text{upper edge}) \quad (2)$$

$$\ell(x, t) = \max\{y : M(x, y, t) = 1\} \quad (\text{lower edge}) \quad (3)$$

The wound area at time t is:

$$A(t) = \sum_{x=0}^{W-1} \max(0, \ell(x, t) - u(x, t) + 1) \quad (4)$$

The biological axiom (monotonic closure) for wound area is a monotonically non-increasing function of time:

$$\boxed{A(t+1) \leq A(t) \quad \forall t \geq 0} \quad (5)$$

Single-Frame Detection

The single-frame detector maps an input image $I(x, y, t)$ to validated edge functions $u_{\text{det}}(x, t)$ and $\ell_{\text{det}}(x, t)$ through a seven-stage pipeline that transforms a raw grayscale image into a geometrically consistent wound mask with smooth boundary curves (Figure 2).

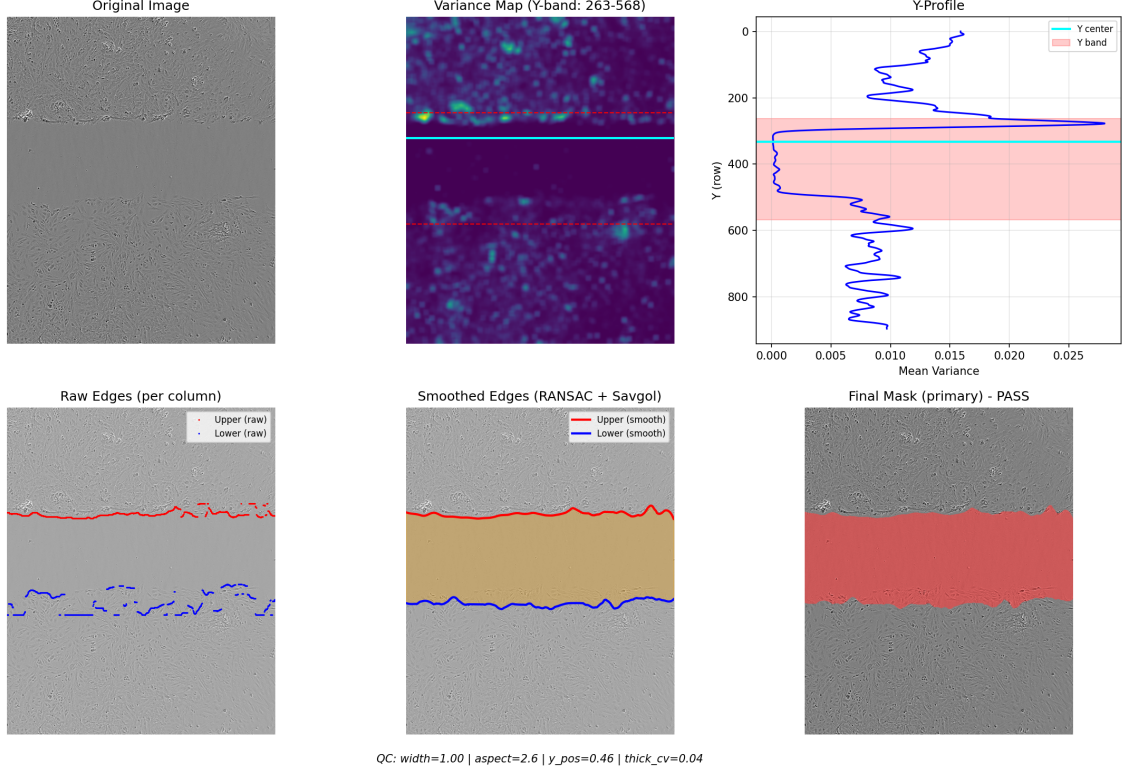


Figure 2: Representative pipeline output on a single frame. Phase-contrast image acquired at $t = 0$ (ALK5i condition, EXP2). The wound channel (center, low local variance) is flanked by cell monolayers (high local variance). Detected upper and lower boundaries are overlaid as colored lines after RANSAC outlier rejection and Savitzky–Golay smoothing. The shaded region corresponds to the reconstructed binary wound mask (Equation (7)).

Stage 1: Local Variance Texture Map. Cell monolayers exhibit high spatial variance due to cell borders, nuclei, and phase-contrast halos, while the wound channel is comparatively smooth [19] Figure 2b). We use local variance as a texture discriminator [23–25]. For a square window of side length w (default $w = 15$), the local variance at each pixel is computed via the uniform-filter identity:

$$\sigma^2(x, y) = \langle I^2 \rangle_w(x, y) - [\langle I \rangle_w(x, y)]^2 \quad (6)$$

where $\langle \cdot \rangle_w$ denotes the box-filter mean over a $w \times w$ neighborhood, yielding $O(HW)$ complexity independent of window size.

Stage 2: Y-Band Detection. The wound is identified as a horizontal band spanning nearly the full image width. We compute the row-averaged variance profile $\bar{V}(y)$, smooth it with a 1-D Gaussian kernel ($\sigma_y = 10.0$ pixels), locate the global minimum y_c , and extract the contiguous segment of sub-threshold rows (35th percentile) containing y_c , expanded by a margin fraction of 0.15.

Stage 3: Column-Wise Edge Extraction. Within the Y-band, upper and lower wound boundaries are extracted for each column. The edge-detection threshold τ is derived adaptively from the cell regions flanking the wound: $\tau = 0.5 \cdot \text{median}(\sigma_{\text{cell}}^2)$. Edge search is fully vectorized using array boolean operations.

Stage 4: RANSAC Outlier Rejection. We apply an iterative polynomial-fit-and-reject scheme inspired by RANSAC [26]. A degree-2 polynomial captures genuine wound curvature while rejecting localized spikes. Three iterations with a maximum deviation threshold of 25 pixels are used.

Stage 5: Savitzky–Golay Smoothing. Residual high-frequency jitter is removed with a Savitzky–Golay filter [27], preserving edge shape without phase distortion. Parameters: window length $w_{\text{sg}} = 51$, polynomial order $p_{\text{sg}} = 3$.

Stage 6: Mask Reconstruction. The binary mask is reconstructed from smooth edges using vectorized broadcasting:

$$M(x, y) = [u(x) \leq y \leq \ell(x)] \quad (7)$$

Stage 7: Geometric Validation. Four quality-control criteria are evaluated on each mask (Supplementary Table S14). A detection passes QC if and only if all four criteria are satisfied. When the primary pipeline fails, an optional fallback strategy is activated using constrained region growing from a seed at y_c with morphological cleanup (hole filling at 500 px, small-object removal at 1000 px, disk closing at radius 5).

Temporal Constraint: Kalman-Filtered Monotonic Closure

A naive implementation of the monotonic closure constraint enforces a hard per-column clamp, such that the upper edge $u(x, t)$ can only move inward relative to $u(x, t - 1)$ (and symmetrically for the lower edge). While this deterministic rule strictly encodes the biological axiom (Equation 5), it introduces an irreversible failure mode: spurious inward detections—arising from debris, illumination artifacts, or transient focus shifts—are permanently propagated to all subsequent frames, with no mechanism for recovery.

To mitigate this limitation, we replace the hard clamp as the primary update mechanism with a per-column Kalman filter. The filter fuses the detector output with a monotonic prior, weighting

each according to a data-driven estimate of observation uncertainty. Noisy or low-contrast detection are down-weighted, while consistent high-confidence observations are retained.

A hard monotonic constraint is subsequently applied as a secondary safety mechanism to guarantee $A(t) \leq A(t-1)$ under all conditions. With this hierarchical design the Kalman filter handles typical artifact-contaminated observations, while the hard constraint provides an unconditional correctness for residual failure cases.

The full per-frame processing sequence of the temporal integrator, including detector input, filtering steps, constraint enforcement, and numerical safeguards, is summarized in Supplementary Table S3. Four numerical safeguards (Supplementary Table S3, Steps 2, 7, 9, 11) prevent failure-mode propagation across frames; their formal statement follows below.

At each timepoint t , the single-frame detector (Section) produces per-column edge estimates $u_{\text{det}}(x, t)$ and $\ell_{\text{det}}(x, t)$, corresponding to the upper and lower wound boundaries, respectively. These estimates are treated as observations and are mapped by the temporal integrator to constrained edges $u(x, t)$ and $\ell(x, t)$ that satisfy the monotonic closure constraint. We denote by $P(x, t)$ the per-column uncertainty associated with the temporal estimate.

At the initial timepoint $t = 0$, the detector outputs are accepted without constraint:

$$u(x, 0) = u_{\text{det}}(x, 0), \quad \ell(x, 0) = \ell_{\text{det}}(x, 0) \quad (8)$$

These edges define the reference wound geometry for the trajectory. Initial per-column uncertainty $P(x, 0)$ is estimated from the spatial residuals between the RANSAC-rejected and Savitzky–Golay smoothed edges at $t = 0$; derivation, fallback values, and numerical safeguards are detailed in Supplementary Table S15.

Kalman-filtered constraint ($t > 0$). For subsequent timepoints, the temporal integrator updates the constrained edges by combining the detector observations with the previous state under a monotonicity prior. The update proceeds in three steps.

Prediction. The predicted state is the previous constrained edge; predicted uncertainty grows by process noise Q :

$$\hat{u}^-(x, t) = u(x, t-1), \quad P^-(x, t) = P(x, t-1) + Q \quad (9)$$

Process noise Q (px²) is estimated empirically from QC-validated trajectories $\mathcal{C}$ as the variance

of consecutive frame-to-frame edge differences:

$$\hat{Q} = \text{Var}\left(\{u_{\text{det}}(x, t+1) - u_{\text{det}}(x, t) \mid x, t \in \mathcal{C}\}\right) \quad (10)$$

A conservative default $Q_0 = 400 \text{ px}^2$ is used on first run.

Spatial pre-smoothing. The raw detected edge is smoothed with a 1-D Gaussian kernel ($\sigma_{\text{spatial}} = 5$ columns) before the Kalman update to suppress column-level artifacts:

$$\tilde{u}_{\text{det}}(x, t) = (G_{\sigma} * u_{\text{det}}(\cdot, t))(x) \quad (11)$$

Per-column observation noise. Observation noise $R(x, t)$ is estimated from three data-driven signals:

$$R(x, t) = R_{\text{base}}(t) \cdot C^{-2}(x, t) \cdot J(x, t) \cdot m_{\text{method}} \quad (12)$$

where $R_{\text{base}}(t) = \text{Var}_x(u_{\text{det}} - \tilde{u}_{\text{det}})$ is the frame-level detector noise, $C(x, t)$ is the local wound-vs-cell variance contrast (low contrast implies high R), $J(x, t)$ is a jump penalty for large deviations from the prediction, and $m_{\text{method}} \in \{1.0, 3.0\}$ penalises the less reliable fallback detector.

Kalman update. The gain $K(x, t) \in [0, 1]$ blends prediction and observation:

$$K(x, t) = \frac{P^-(x, t)}{P^-(x, t) + R(x, t)} \quad (13)$$

$$u_{\text{filt}}(x, t) = u(x, t-1) + K(x, t)[\tilde{u}_{\text{det}}(x, t) - u(x, t-1)] \quad (14)$$

$$P_{\text{upd}}(x, t) = (1 - K(x, t)) \cdot P^-(x, t) \quad (15)$$

High R (noisy observation) implies small K , causing the filter to rely on the monotonic prior. Low R (clean, high-contrast frame) implies large K , trusting the detector.

Hard safety net (secondary guarantee). After the Kalman blend, the biological axiom is enforced unconditionally:

$$u(x, t) = \max(u_{\text{filt}}(x, t), u(x, t-1)) \quad (16)$$

$$\ell(x, t) = \min(\ell_{\text{filt}}(x, t), \ell(x, t-1)) \quad (17)$$

This guarantees $A(t) \leq A(t-1)$ for all $t > 0$ regardless of Kalman output.

The pipeline ultimately produces, for each trajectory, a sequence of wound masks and the corresponding area time series $A(t)$. The Kalman process adds $O(W)$ operations per frame. Total per-frame complexity remains $O(HW)$, dominated by the variance map computation.

Wound Closure. When the normalized wound area satisfies $\frac{A(t-1)}{A(0)} < \theta_{\text{close}}$ with $\theta_{\text{close}} = 0.01$ the wound is considered fully closed. For all subsequent timepoints, the edge positions are held constant $u(x, t) = u(x, t-1)$ and $\ell(x, t) = \ell(x, t-1)$, preventing numerical fluctuations after closure.

Wound Quantification

Once the wound boundaries have been extracted and temporally constrained, quantitative measures of wound healing can be derived from the resulting area trajectory $A(t)$ and edge positions $u(x, t)$ and $\ell(x, t)$. These metrics represent the inward movement of the wound boundaries at each column x .

Fractional wound closure is defined as the proportion of the initial wound area that has closed over time. It ranges from 0 (no closure) to 1 (complete closure), and is given by

$$f(t) = \frac{A(0) - A(t)}{A(0)}. \quad (18)$$

To quantify spatial dynamics, we measure the displacement of the upper and lower wound edges between the initial timepoint $t = 0$ and a target time t_T , defining the edge displacement as

$$\Delta u(x) = u(x, t_T) - u(x, 0), \quad (19)$$

$$\Delta \ell(x) = \ell(x, 0) - \ell(x, t_T). \quad (20)$$

Wound healing rate is defined by converting the edge displacements of Equations (19)–(20) into velocities through normalization by elapsed time. The mean closure speed represents the average inward movement of the wound boundaries and is defined as

$$v_{\text{mean}} = \frac{v_{\text{upper}} + v_{\text{lower}}}{2}, \quad (21)$$

where each edge velocity is computed as the average displacement per unit time across the set $\mathcal{V}$ of

valid columns:

$$v_{\text{edge}} = \frac{1}{|\mathcal{V}|} \sum_{x \in \mathcal{V}} \frac{|\Delta_{\text{edge}}(x)|}{t_T}. \quad (22)$$

Averaging over valid columns suppresses localized edge artifacts without discarding the per-column spatial resolution preserved throughout processing.

Statistical Analysis

Pairwise treatment comparisons versus DMSO were performed using Welch’s *t*-test (no equal-variance assumption), stratified by cell line. Multiple comparison correction was applied using the Bonferroni method. Effect sizes were quantified as Cohen’s *d*, defined as $d = \frac{\bar{x}_{\text{treated}} - \bar{x}_{\text{control}}}{s_{\text{pooled}}}$.

The time-series analysis was performed using a linear mixed-effects model. The response variable distance_{ij} for trajectory *i* at timepoint *j* was modeled as

$$\begin{aligned} \text{distance}_{ij} = & \mu + \beta_1 t_j + \beta_2 \text{condition}_i + \beta_3 \text{cell_line}_i \\ & + \beta_4 (t_j \times \text{condition}_i) + \beta_5 (t_j \times \text{cell_line}_i) + b_{0i} + b_{1i} t_j + \varepsilon_{ij}. \end{aligned} \quad (23)$$

Here, μ denotes the fixed intercept, β_k the fixed-effect coefficients, b_{0i} and b_{1i} the random intercept and slope for trajectory *i*, and ε_{ij} the residual error term.

Model assumptions were evaluated by testing residual normality using the Shapiro–Wilk test and homogeneity of variance using Levene’s test (see Supplementary Table S8).

Implementation

The pipeline was implemented in Python 3.11 using NumPy 2.4.2, OpenCV 4.12.0, SciPy 1.16.1, and scikit-image 0.25.2 [28]. The pipeline is fully deterministic and reproducible. The per-frame computational complexity is $O(HW)$, dominated by the variance map computation. For a typical 1024×1280 image, this corresponds to approximately 1.3×10^6 operations per stage.

Results

Validation was performed on 124 wound-bearing phase-contrast images drawn from 134 baseline frames acquired across two cell lines (iMC ISOR544C and iMC MUTR544C) and two independent

experiments (EXP1, $n = 62$; EXP2, $n = 72$). Ten images (7.5%) were excluded for containing no visible wound at $t = 0$; 9 of 10 occurred in EXP1 and 8 of 10 in the iMC MUTR544C line (Supplementary Table S10).

Wound Detection Accuracy

Of 124 wound-bearing images, the pipeline correctly segmented 122 (98.4%) as judged by expert review. Only 2 images were incorrectly segmented, both in the iMC MUTR544C line in EXP1. Performance was consistent across conditions and cell lines (Table 1), and reproducible across experiments (EXP1: 51/53, 96.2%; EXP2: 71/71, 100.0%).

Table 1: Detection accuracy by cell line and condition (wound-bearing images only). Values are expressed as correctly segmented / total (percentage). Expert review served as ground truth.

Cell line	ALK5i	DMSO	Media
Line 1	30/30 (100.0%)	16/16 (100.0%)	15/15 (100.0%)
Line 2	25/26 (96.2%)	30/31 (96.8%)	6/6 (100.0%)
Overall: 122/124 (98.4%)			

Automated Quality-Control Performance

Automated QC achieved 98.5% agreement with the human reviewer across all annotated images ($n = 134$; Cohen’s $\kappa = 0.90$; Supplementary Table S11). The module identified all 122 correctly segmented images (sensitivity = 1.00, NPV = 1.00) and flagged 10 of 12 problematic images as invalid (specificity = 0.83). The two false positives — incorrectly segmented images that passed QC — were indistinguishable from valid segmentations by variance-based scoring alone. Within the wound-bearing cohort ($n = 124$), the pipeline correctly classified 123 of 124 images (accuracy = 99.2%) without false negatives. QC is therefore a safeguard for quantitative conclusions, not a cosmetic post-filter; automated rejection becomes essential when trajectory counts exceed practical limits for manual review.

Comparison with Existing Tools

Supplementary Table S13 contextualises the pipeline against established and recent alternatives. The pipeline proposes a training-free operation with a hard temporal closure constraint, and multi-level

automated quality control.

Wound Closure Dynamics

After QC validation, 59 trajectories with a cross-sectional measurement at the final timepoint entered the wound closure dynamics analysis: DMSO ($n = 23$), Media ($n = 21$), ALK5i ($n = 15$), across two cell lines: iMC ISOR544C ($n = 33$) and iMC MUTR544C ($n = 26$). Each trajectory comprised 20–25 timepoints (Figure 3).

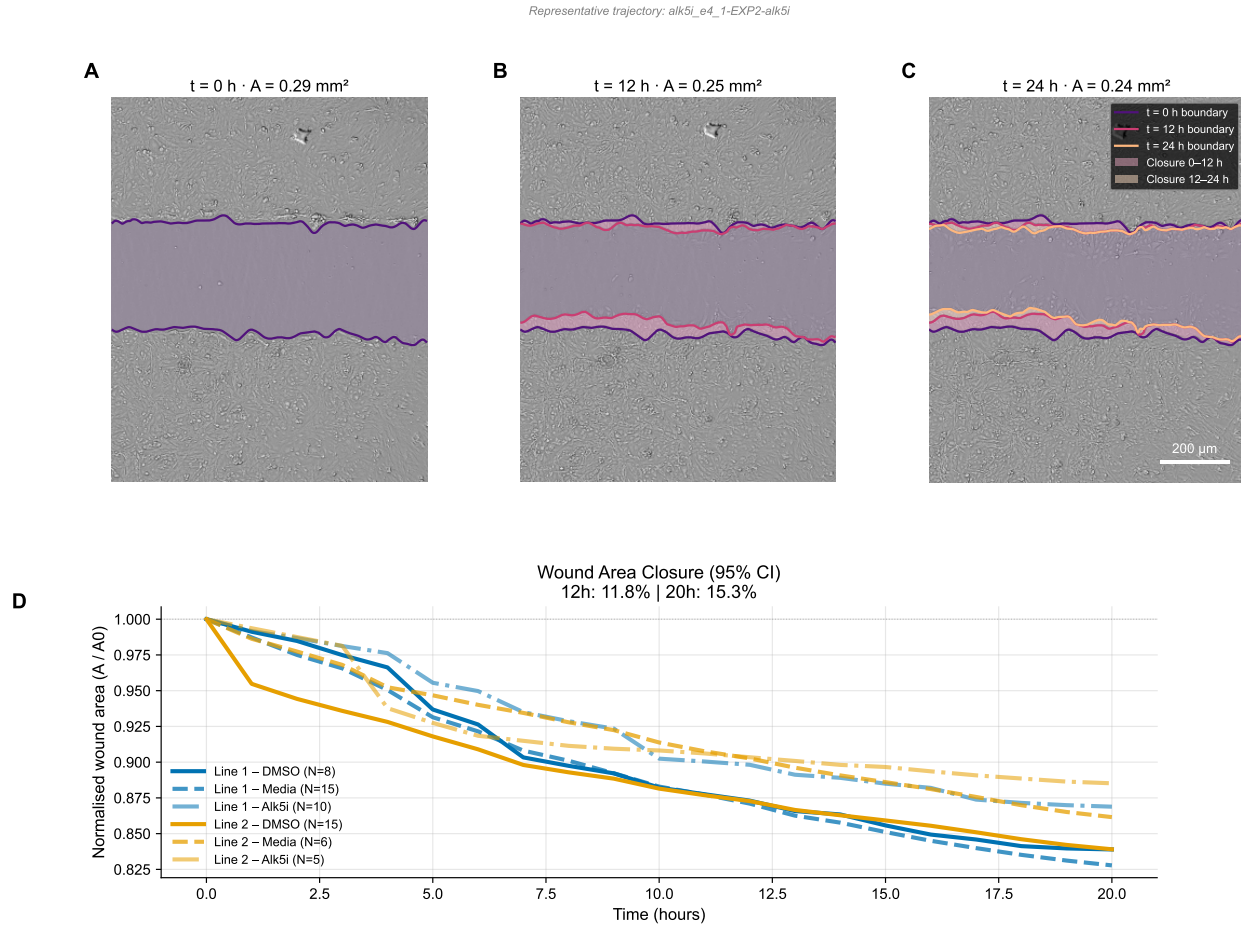


Figure 3: Automated wound boundary detection and model quality metrics. (A–C) Representative phase-contrast images of a wound healing assay at three timepoints ($t = 0$, mid, and $t = 20 \text{ h}$) with computationally detected wound boundaries overlaid as semi-transparent filled regions. Earlier timepoints are shown as faint dashed outlines; the current timepoint as solid fill. Images were acquired at $1.24 \mu\text{m}/\text{px}$. (D) Normalized wound area over time, expressed as a fraction of the $t = 0$ wound area. Error bands represent 95% confidence intervals. Sample sizes per group are provided in the legend. (Figure 3 illustrates a single representative cell line/condition subset; the full dataset is shown in Figure 4.)

By construction, the temporal constraint produced strictly non-increasing wound-area trajectories

as depicted in Figure 3D and in Supplementary Table S5. Across the 59 QC-validated trajectories, mean normalized closure reached 11.8% ($\pm 0.8\%$ SEM) at 12 hours and 15.3% ($\pm 0.9\%$ SEM) at 20 hours, with the ALK5i condition lowest at both timepoints (Supplementary Table S12).

Retaining the six trajectories that failed $t = 0$ QC — four in the ALK5i group in the iMC MUTR544C line — introduced spurious masks that inflated apparent ALK5i closure to 31.0% ($\pm 8.5\%$ SEM) and inverted the condition ranking relative to the edge-displacement metric. Restricting to QC-validated trajectories removed this artifact and reconciled the area-based and displacement-based metrics.

Wound closure trajectories followed a characteristic pattern across all conditions (Figure 4): an initial lag phase (0–4 h) with minimal closure, followed by progressive inward displacement that did not plateau within 24 hours. Per-condition mean trajectories separated with non-overlapping 95% confidence bands from 12 h onward, demonstrating that the pipeline resolves condition-dependent closure differences at the scale of this cohort.

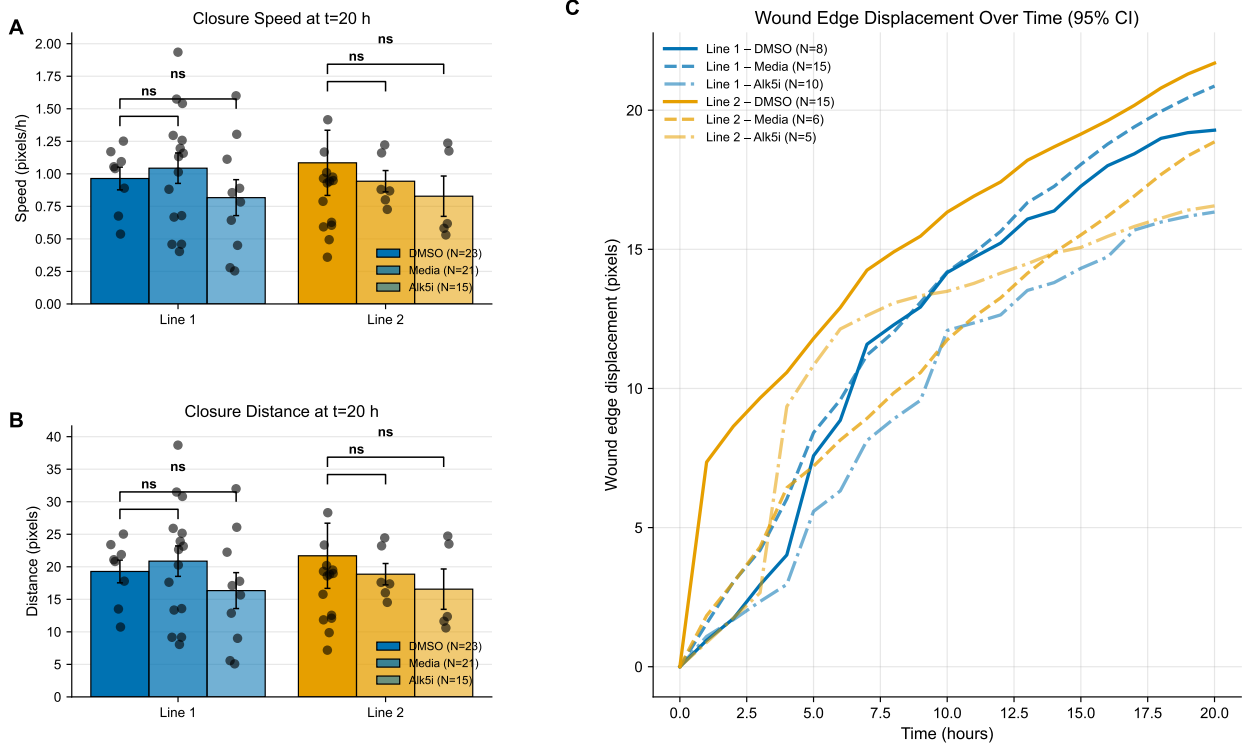


Figure 4: Wound closure dynamics across experimental conditions. (A) Mean wound closure speed (pixels/h) at $t = 20$ h, computed as the rate of wound edge displacement. (B) Mean wound closure distance (pixels) at $t = 20$ h, representing cumulative inward displacement of the wound boundary from $t = 0$. (C) Time-series of wound edge displacement over the full observation period. Error bars represent 95% confidence intervals. Statistical comparisons were performed using Welch's t -test with Bonferroni correction. Sample sizes: Line 1 – ALK5i: $n = 10$; Line 1 – DMSO: $n = 8$; Line 1 – Media: $n = 15$; Line 2 – ALK5i: $n = 5$; Line 2 – DMSO: $n = 15$; Line 2 – Media: $n = 6$.

Cross-sectional analysis at the final timepoint yielded no statistically significant pairwise differences after Bonferroni correction across both cell lines (Line 1: ALK5i vs. DMSO $p = 0.38$, $d = -0.40$; Line 2: $p = 0.40$, $d = -0.29$; full results in Supplementary Table S6). Media did not differ from DMSO in either cell line ($p > 0.59$). Corrected p -values of 1.000 reflect per-group sample sizes of $n = 5$ –15 and a mean closure range of $\sim 20\%$ over 24 h, not a failure of the pipeline to resolve differences of this magnitude.

A linear mixed-effects model (Equation (23)) confirmed a significant main effect of time ($\beta = 0.880$ px/frame, $p < 10^{-12}$), establishing that wound closure proceeded consistently across all conditions. No time $\times$ condition or time $\times$ cell line interaction reached significance (all $p > 0.31$), consistent with the cross-sectional findings. Full sample accounting, QC pass rates, descriptive statistics, pairwise comparisons with confidence intervals, and inter-experiment reproducibility are

provided in Supplementary Tables S1–S7.

Discussion

We report a fully automated, training-free pipeline for scratch-wound healing quantification that achieves 98.4% detection accuracy across 134 phase-contrast images without annotated training data, parameter tuning, or specialist infrastructure. Although scratch-wound assays have been performed for decades, no existing tool enforces the physical reality that a wound can only close over time: frame-independent approaches generate area trajectories with non-biological fluctuations that require post-hoc correction before statistical analysis can proceed [20]. The presented pipeline addresses this gap by coupling texture-based single-frame detection with a biologically grounded monotonic closure constraint, and is released as open-source software [1] requiring only NumPy, OpenCV, and SciPy on standard laboratory hardware.

The single-frame detector is grounded in a first-principles observation: cells scatter light and produce high local variance in phase-contrast images, while the cell-free wound channel does not [19]. This texture contrast is a stable discriminator that eliminates reliance on annotated training sets and the domain-shift risk that accompanies deep-learning models applied outside their training distribution [29, 30]. The detection accuracy achieved without any supervision is competitive with supervised methods on comparable assays, and the two-stage boundary regularization — RANSAC-inspired outlier rejection followed by Savitzky–Golay smoothing — provides principled separation between genuine wound curvature and artifact-driven edge spikes [26, 27].

The monotonic closure constraint encodes a biological axiom of the assay: wound area is non-increasing over time. Enforcing this per column — strictly stronger than a global area constraint — prevents non-biological area re-openings that would otherwise propagate into erroneous closure-rate estimates [20]. The Kalman-filtered formulation refines this further: rather than clamping edges deterministically, the filter down-weights low-confidence observations arising from debris or illumination artifacts, allowing the trajectory to recover from transient detection failures while the hard safety net guarantees the biological axiom under all conditions. The extracted trajectories supported a linear mixed-effects model with a significant closure-over-time main effect ($\beta = 0.88$ px/frame, $p < 10^{-12}$; Equation (23)), consistent with the distributional and stationarity requirements for linear

mixed-effects inference. Cross-sectional pairwise comparisons between conditions yielded corrected p -values of 1.000, consistent with the modest closure range ($\sim 20\%$ in 24 h) and small per-group sample sizes of this test bed; these results reflect the limitations of the experimental design rather than a failure of detection.

The automated quality control from the QC module achieved a sensitivity and NPV of 1.00, ensuring no valid segmentation is discarded. Exclusion of the six QC-failing trajectories removed spurious ALK5i masks that would otherwise have inverted the area-based closure ranking relative to the edge-displacement metric. QC is therefore a safeguard for quantitative conclusions when manual review is not feasible.

The approach has limitations that should be acknowledged. The monotonic constraint carries an irreversibility cost at the per-column level: a sufficiently large spurious inward detection, from a bubble or severe focus shift, will propagate to later frames even under Kalman filtering. The edge-detection threshold assumes unimodal cell variance and may require adaptation for heterogeneous populations or images with illumination gradients. Edges are resolved at integer-pixel resolution, introducing quantization error as wounds approach closure, and all measurements are reported in pixels; incorporation of pixel-size metadata would yield physically calibrated areas and speeds. Pixel-level validation rested on five annotated images, and a larger ground-truth dataset would tighten estimates of spatial accuracy across imaging conditions and cell types.

Future development will address sub-pixel edge localization via parabolic fitting, adaptive per-column thresholding, and bootstrap-based uncertainty bounds for area measurements. The strictly monotonic, smooth trajectories produced by this pipeline are also well-conditioned inputs for fitting mechanistic models of collective migration — in particular Fisher–Kolmogorov reaction-diffusion frameworks [31–33] — enabling direct estimation of cell motility and proliferation rates from the same trajectory data.

On this test bed, the pipeline produced temporally consistent wound-area trajectories across 134 phase-contrast images and two cell lines, with 98.4% detection accuracy and a mean Dice coefficient of 0.91 against expert annotations. The automated quality-control system correctly excluded spurious segmentations that would have otherwise inverted the condition ranking, and the resulting trajectories supported well-specified downstream statistical inference. The approach requires no training data, no parameter tuning, and runs in linear time per frame on standard

laboratory hardware, offering a practical improvement over frame-independent tools for this class of phase-contrast scratch-wound assays.

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Supplementary Material

Supplementary Tables

Table S1: Sample sizes (number of trajectories) per cell line and treatment condition after QC filtering.

Cell line	ALK5i	DMSO	Media
Line 1	10	8	15
Line 2	5	15	6

Table S2: Sample accounting: initial, QC-passing, and final trajectory counts per group, with exclusion percentages. Exclusions reflect trajectories that failed $t = 0$ quality control; all QC-passing trajectories were retained for analysis.

Cell line	Condition	n initial	n QC pass	n final	n excluded	Excl. (%)
Line 1	ALK5i	11	10	10	1	9.1
Line 1	DMSO	8	8	8	0	0.0
Line 1	Media	15	15	15	0	0.0
Line 2	ALK5i	8	5	5	3	37.5
Line 2	DMSO	16	15	15	1	6.2
Line 2	Media	7	6	6	1	14.3

Table S3: Per-frame processing sequence of the temporal integrator ($t > 0$).

Step	Operation	Implemented in
1	Single-frame detector $\rightarrow$ raw edges $u_{\text{det}}(x, t)$, $\ell_{\text{det}}(x, t)$	Section 2.3.2
2	NaN guard: replace non-finite detector outputs with previous state	<code>KalmanEdgeFilter.update</code>
3	Spatial Gaussian smoothing of detected edges ($\sigma_{\text{spatial}} = 5$ columns)	<code>gaussian_filter1d</code>
4	Per-column observation noise $R(x, t)$ estimation	<code>estimate_R_per_column</code>
5	Kalman predict step: $P^-(x, t) = P(x, t - 1) + Q$	<code>_kalman_step</code>
6	Kalman update: gain K , blended edge u_{flt} , contracted P_{upd}	<code>_kalman_step</code>
7	NaN guard on Kalman output: reset non-finite u_{flt} to $u(x, t - 1)$	<code>KalmanEdgeFilter.update</code>
8	Hard safety net: $u(x, t) = \max(u_{\text{flt}}, u(x, t - 1))$	<code>KalmanEdgeFilter.update</code>
9	P-state correction: reset P to P^- for columns where safety net fired	<code>KalmanEdgeFilter.update</code>
10	Mask reconstruction from constrained edges	<code>_edges_to_mask</code>
11	Last-resort area backstop: emit warning and revert if $A(t) > A(t - 1)$	<code>QuantificationSegmenter.segment</code>

Table S4: Descriptive statistics for the primary endpoint (wound closure distance, pixels) and secondary endpoint (closure speed, pixels/h) at the final timepoint, stratified by cell line and treatment condition. Values are mean $\pm$ SEM with 95% confidence intervals.

Cell line	Condition	Distance mean	Distance SD	Distance SEM	CI ₉₅ lower	CI ₉₅ upper	<i>n</i>
Line 1	ALK5i	16.34	8.73	2.76	10.93	21.75	10
Line 1	DMSO	19.28	4.94	1.75	15.86	22.70	8
Line 1	Media	20.87	9.06	2.34	16.28	25.45	15
Line 2	ALK5i	16.56	6.93	3.10	10.49	22.64	5
Line 2	DMSO	21.69	19.40	5.01	11.87	31.51	15
Line 2	Media	18.86	4.02	1.64	15.65	22.08	6

Table S5: QC pass rate and monotonic constraint rate per group. All QC-passing trajectories were constrained, consistent with the biological axiom.

Cell line	Condition	QC pass rate (%)	Constraint rate (%)	<i>n</i> total
Line 1	ALK5i	90.9	100	11
Line 1	DMSO	100.0	100	8
Line 1	Media	100.0	100	15
Line 2	ALK5i	62.5	100	8
Line 2	DMSO	93.8	100	16
Line 2	Media	85.7	100	7

Table S6: Pairwise comparisons of closure speed and distance versus DMSO, stratified by cell line. Welch’s *t*-test; *p*-values shown before and after Bonferroni correction (4 tests per metric). All corrected *p*-values equal 1.000.

Cell line	Condition	Metric	<i>t</i> -stat	Cohen’s <i>d</i>	<i>p</i> (raw)	<i>p</i> (Bonf.)	<i>n</i> _{trt}	<i>n</i> _{ctrl}
Line 1	Media	speed	0.543	0.200	0.593	1.000	15	8
Line 1	ALK5i	speed	−0.900	−0.402	0.382	1.000	10	8
Line 2	Media	speed	−0.537	−0.169	0.599	1.000	6	15
Line 2	ALK5i	speed	−0.871	−0.294	0.396	1.000	5	15
Line 1	Media	distance	0.543	0.200	0.593	1.000	15	8
Line 1	ALK5i	distance	−0.900	−0.402	0.382	1.000	10	8
Line 2	Media	distance	−0.537	−0.169	0.599	1.000	6	15
Line 2	ALK5i	distance	−0.871	−0.294	0.396	1.000	5	15

Table S7: Mixed-effects model parameter estimates. Model: distance $\sim$ time $\times$ condition $\times$ cell_line, with random intercept and random slope per trajectory. The main effect of time is highly significant ($p < 10^{-12}$); no interaction terms reach significance.

Term	Estimate	p -value	CI ₉₅ lower	CI ₉₅ upper
Intercept	0.924	0.744	−4.618	6.467
DMSO vs. ALK5i	0.787	0.853	−7.527	9.101
Media vs. ALK5i	1.301	0.722	−5.855	8.456
Line 2 vs. Line 1	2.984	0.542	−6.616	12.584
DMSO $\times$ Line 2	1.958	0.755	−10.332	14.248
Media $\times$ Line 2	−3.255	0.618	−16.056	9.545
t (main effect)	0.880	<0.001	0.639	1.122
$t \times$ DMSO	0.142	0.442	−0.220	0.505
$t \times$ Media	0.161	0.313	−0.151	0.473
$t \times$ Line 2	−0.115	0.591	−0.533	0.304
$t \times$ DMSO $\times$ Line 2	−0.069	0.801	−0.605	0.467
$t \times$ Media $\times$ Line 2	−0.024	0.932	−0.582	0.534
<i>Variance components</i>				
Group variance	9.016	<0.001	5.434	12.598
Group $\times t$ covariance	0.104	0.067	−0.007	0.215
t variance	0.016	<0.001	0.009	0.023

Table S8: Statistical assumption checks. Shapiro–Wilk normality test for wound closure distance, per group. Levene’s test for homogeneity of variance across conditions, per cell line. DMSO Line 2 shows significant deviation from normality ($p < 0.001$), warranting use of Welch’s t -test (unequal variances assumed).

Cell line	Condition	Shapiro W	Shapiro p	n
Line 1	ALK5i	0.961	0.799	10
Line 1	DMSO	0.915	0.393	8
Line 1	Media	0.957	0.634	15
Line 2	ALK5i	0.780	0.055	5
Line 2	DMSO	0.551	<0.001	15
Line 2	Media	0.876	0.253	6
<i>Levene’s test (homogeneity of variance across conditions)</i>				
Line 1	$F = 1.457, p = 0.249$			
Line 2	$F = 0.412, p = 0.667$			

Table S9: Inter-experiment reproducibility: mean wound closure distance (pixels) at the final timepoint, by cell line, condition, and experiment. Experiment 2 data were unavailable for some condition–cell line combinations (see text for details).

Cell line	Condition	Mean EXP1	Mean EXP2	Difference	<i>n</i> EXP1	<i>n</i> EXP2
Line 1	ALK5i	10.94	17.69	-6.75	2	8
Line 1	DMSO	19.28	—	—	8	0
Line 1	Media	17.63	23.70	-6.07	7	8
Line 2	ALK5i	16.56	—	—	5	0
Line 2	DMSO	25.82	16.98	8.84	8	7
Line 2	Media	18.86	—	—	6	0

Table S10: Distribution of no-wound images by cell line and condition. These 10 images contained no visible scratch wound at $t = 0$ and were excluded from the accuracy analysis cohort.

Cell line	ALK5i	DMSO	Media	Total
Line 1	2	0	0	2
Line 2	6	1	1	8
Total	8	1	1	10

Table S11: QC module confusion matrix against expert review (all annotated images, $n = 134$). Sensitivity = 1.00; Specificity = 0.83; PPV = 0.98; NPV = 1.00; accuracy = 0.99.

	Human: Correct	Human: Incorrect
QC: Valid	122 (TP)	2 (FP)
QC: Invalid	0 (FN)	10 (TN)

Table S12: Mean fractional wound closure by condition at 12 h and 20 h, pooled across cell lines (mean $\pm$ SEM; $n = 59$ QC-validated trajectories). ALK5i shows the lowest closure at both timepoints.

Condition	<i>n</i>	Closure 12 h (%)	SEM	Closure 20 h (%)	SEM
ALK5i	15	10.0	1.2	12.6	1.5
DMSO	23	12.7	1.7	16.1	1.7
Media	21	12.0	1.1	16.3	1.5

Table S13: Comparison of wound-healing quantification tools. DL, deep learning; RQSA, Robust Quantitative Scratch Assay.

Feature	TScratch [9]	ImageJ [10]	RQSA [11]	DL [30]	Level-set [14]	This work
Training data	No	No	No	Yes	No	No
Temporal constraint	No	No	No	No	No	Yes
Automated QC	No	No	Partial	No	No	Yes (3-level)
Spatial analysis	No	Width only	No	Pixel-level	Sector	Per-column
Outlier rejection	No	No	Statistical	Implicit	No	RANSAC
Parallel processing	No	Macro	No	GPU batch	No	CPU parallel

Table S14: Geometric validation criteria applied to each wound mask at Stage 7 of the single-frame detector. A detection passes QC if and only if all four criteria are satisfied simultaneously.

Criterion	Threshold	Rationale
Width coverage r_w	> 0.75	Wound must span most of image width
Aspect ratio ρ	> 1.5	Wound is wider than tall
Y-position $\bar{y}_{\text{frac}}$	$\in (0.15, 0.85)$	Wound is roughly centered vertically
Thickness CV	< 0.60	Consistent thickness across columns

Table S15: Pipeline parameter values and Kalman initialisation details. All parameters are fixed across experiments and cell lines; none require user tuning. $P(x, 0)$ initialisation: initial uncertainty is estimated as $P(x, 0) = \max(R_{\text{base}}(0), 1.0)$, where $R_{\text{base}}(0) = \text{Var}_x(u_{\text{RANSAC}}(x, 0) - u_{\text{det}}(x, 0))$ is the within-frame spatial residual variance between RANSAC-rejected and Savitzky–Golay smoothed edges at $t = 0$. Fallback: if fewer than 6 finite-valued columns are available or fallback detection is active at $t = 0$, $R_{\text{base}}(0) = 0.1 \times Q_0 = 40 \text{ px}^2$. The floor of 1.0 prevents zero initial uncertainty from collapsing the Kalman gain. Numerical safeguards: (i) non-finite R clamped to Q_0 ; (ii) non-finite Kalman output reset to previous edge; (iii) P reset to P^- when hard safety net fires; (iv) last-resort revert if $A(t) > A(t - 1)$ persists after all constraint paths.

Stage	Parameter	Value	Purpose
Variance map	Window size w	15 px	Local texture estimation
Y-band	Row-profile Gaussian σ_y	10.0 px	Wound centre smoothing
Y-band	Variance percentile threshold	35th	Sub-threshold band selection
Y-band	Margin expansion fraction	0.15	Band boundary padding
Edge extraction	Threshold factor	0.5	$\tau = 0.5 \cdot \text{median}(\sigma_{\text{cell}}^2)$
RANSAC	Polynomial degree	2	Captures genuine curvature
RANSAC	Iterations	3	Outlier convergence
RANSAC	Max deviation	25 px	Spike rejection threshold
Savitzky–Golay	Window length w_{sg}	51	High-frequency jitter removal
Savitzky–Golay	Polynomial order p_{sg}	3	Shape preservation
Fallback detection	Morphological hole fill	500 px	Region growing cleanup
Fallback detection	Small-object removal	1000 px	Region growing cleanup
Fallback detection	Disk closing radius	5 px	Region growing cleanup
Kalman	Default process noise Q_0	400 px^2	Conservative first-run default
Kalman	Spatial pre-smoothing σ_{spatial}	5 columns	Column-level artifact suppression
Kalman	Fallback R clamp	Q_0	Non-finite guard
Wound closure	Closure threshold θ_{close}	0.01	Normalised area floor